

Prolonged N-acetylcysteine therapy in late acetaminophen poisoning associated with acute liver failure--a need to be more cautious?

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Since the 1970s, N-acetylcysteine (NAC) has shown proven efficacy as an antidote for acetaminophen (APAP) poisoning and APAP-induced liver failure for early presenters. The current evidence of benefits of NAC for late presenters is controversial because of the poor understanding of the mechanism of late toxicity. In the previous issue of Critical Care, Yang and colleagues use a mouse model to demonstrate that NAC in doses similar to those used therapeutically to treat APAP poisoning in humans impairs liver regenerative capacity and that the effect is more pronounced when administered for a longer duration. Studies based on cell cultures support this evidence. Cytokine and growth factor signalling pathways are recognised to be involved in the process of liver regeneration and apoptosis. This research paper generates several issues related to the future management of APAP-induced liver failure and research into the mechanism of toxicity, especially of late toxicity.

Intravenous acetylcysteine for the treatment of acetaminophen overdose.

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Acetaminophen is a leading cause of overdose-related hepatotoxicity. Although acetylcysteine prevents or minimizes acetaminophen-induced hepatotoxicity and reduces mortality, some patients presenting with complicated overdose scenarios (massive ingestions or combination or modified-release formulations) may develop toxicity despite administration of recommended dosage regimen. The article evaluates evidence regarding intravenous acetylcysteine's effectiveness in patients with acute overdoses who receive treatment within 10 h or > 10 h, patients with chronic supratherapeutic ingestions, and those with acetaminophen-induced fulminant hepatic failure. Intravenous and oral acetylcysteine are compared. A one-size-fits-all approach towards acetylcysteine therapy provides suboptimal care in some patients. High-risk patients are identified. Specific discontinuation criteria are presented. The standard intravenous regimen will effectively treat most early-presenting uncomplicated overdoses. Acetylcysteine dosing should be individualized in patients with complicated presentations and in particular situations in which plasma acetaminophen concentrations may be persistently elevated at the end of the infusion or in late presenters. More studies are needed to evaluate the optimal intravenous dosage regimen and the role of oral acetylcysteine in these high-risk patients. Treatment decisions may be aided by consultation with a poison center and/or clinical toxicologist.

Plasma paracetamol concentration at hospital presentation has a dose-dependent relationship with liver injury despite prompt treatment with intravenous acetylcysteine.

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Paracetamol (acetaminophen) overdose is a common reason for emergency hospital admission in the UK and the leading cause of acute liver failure in the Western world. Currently, the antidote acetylcysteine (NAC) is administered at a dose determined only by body weight without regard for the body burden of paracetamol. To determine whether higher plasma paracetamol concentrations are associated with increased risk of liver injury despite prompt treatment with intravenous NAC. Patients admitted to

hospital for treatment with intravenous NAC following a single acute paracetamol overdose entered the study if NAC was commenced within 24 h of drug ingestion (N = 727 hospital presentations). Based on the plasma paracetamol concentration at first presentation to hospital, a series of nomograms were created: 0-100, 101-150, 151-200, 201-300, 301-500 and over 501 mg/L. The primary endpoints were acute liver injury (ALI - peak serum ALT activity >150 U/L and double the admission value) and hepatotoxicity (peak ALT >1000 U/L). ALI and hepatotoxicity were more common in patients with higher admission plasma paracetamol concentrations despite NAC treatment (ALI: nomogram 0-100: 6%, 101-150: 3%, 151-200: 3%, 201-300: 9%, 301-500: 13%, over 501 mg/dL: 27%. $p < 0.0001$). This dose-response relationship between paracetamol concentration and ALI persisted even in patients treated with NAC within 8 h of overdose (nomogram 0-100: 0%, 101-150: 0.8%, 151-200: 2%, 201-300: 3.6%, 301-500: 12.5%, over 501 mg/L: 33%. $p < 0.0001$) and in patients with normal ALT activity at first presentation (nomogram: 0-100: 0%, 101-150: 1.2%, 151-200: 1.5%, 201-300: 5.3%, 301-500: 10.8% $p < 0.0001$). Patients with increased concentrations of plasma paracetamol at hospital presentation are at higher risk of liver injury even when intravenous NAC is promptly administered before there is biochemical evidence of toxicity. This study supports theoretical concerns that the current intravenous dose of NAC may be too low in the setting of higher paracetamol exposure.

Intravenous N-acetylcysteine for paracetamol poisoning.

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Intravenously administered N-acetylcysteine was used to treat 11 patients at risk of hepatic damage from paracetamol poisoning. Complete protection against liver failure was seen in all cases without any adverse effects. The efficacy and safety of intravenously administered N-acetylcysteine make it the treatment of choice in paracetamol poisoning.

Large paracetamol overdose - Higher dose NAC is NOT required.

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Paracetamol overdose is common in developed countries but less than 10% involve large ingestions exceeding 30 g or 500 mg/kg. High dose acetylcysteine (NAC) has been proposed in patients taking large paracetamol overdoses based on reports of hepatotoxicity despite early initiation of NAC treatment with the commonly used 300 mg/kg intravenous acetylcysteine regimen. The evidence from cohorts of patients treated with the standard NAC regimen after large paracetamol overdoses shows that it is effective in most patients. A small study in patients with paracetamol overdoses of 40 g or more and paracetamol concentrations above the 300 mg/L nomogram line showed that modification of the standard NAC regimen to provide a total of 400-500 mg/kg NAC over 21-22 h may reduce the risk of hepatotoxicity (peak ALT >1000 IU/L) but the impact on development of hepatic failure, liver transplantation and mortality with this approach is presently unknown. Better risk stratification of patients taking paracetamol overdose may allow higher dose NAC and adjunctive treatments such as CYP2E1 inhibition and extracorporeal removal of paracetamol to be targeted to those patients at the highest risk of hepatotoxicity after a large paracetamol overdose.

[Paracetamol overdose: a new form of suicide in Chile and the value of N-acetylcysteine administration].

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Overdose of paracetamol is associated to varying degrees of hepatic necrosis and may lead to severe hepatic failure. In this paper, we report 2 patients with paracetamol overdose for suicidal purposes who were successfully treated using the antidote N-acetyl-cysteine orally. A brief analysis of the literature follows.

Fomepizole Therapy for Acetaminophen-Induced Liver Failure in an Infant.

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Acetaminophen overdose is common in the pediatric population. N-acetylcysteine (NAC) is effective at preventing liver injury in most patients when started shortly after the overdose. Delays to therapy increase risk of hepatotoxicity and liver failure that may necessitate organ transplant. Animal studies have demonstrated fomepizole may provide added benefit in acetaminophen overdose because of its ability to block the metabolic pathway that produces the toxic acetaminophen metabolite and downstream inhibition of oxidative stress pathways that lead to cell death. Several adult case reports describe use of fomepizole in patients at higher risk for poor outcomes despite NAC. We describe a case of a 7-month-old female who presented in acute liver failure with persistently elevated acetaminophen concentration secondary to repeated supratherapeutic doses of acetaminophen to manage fever. Fomepizole and NAC antidotes were used in the management of the patient. She fully recovered despite demonstrating multiple markers of poor outcome on initial presentation. Although randomized trials are lacking, this case suggests that fomepizole may safely provide additional benefit in pediatric patients at risk for severe acetaminophen toxicity.

A Case of Cocaine-Induced Acute Liver Failure Reversed With N-Acetylcysteine.

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Acute liver failure (ALF) is a life-threatening injury that is most often caused by drug-induced injury, including acetaminophen overdose, in the United States. The hallmarks of ALF are hepatic encephalopathy and coagulopathy in a patient without an established history of liver disease. While acetaminophen overdose has an antidote, that is N-acetylcysteine (NAC), when given acutely, most other causes of hepatic failure require an urgent liver transplant. In this paper, we report a case of cocaine-induced acute liver failure that was reversed with the administration of NAC. Our case began when a middle-aged male presented to the emergency department complaining of nausea, vomiting, fatigue, and confusion for the past three days. His past medical history was pertinent for a history of opioid use disorder and his physical exam was remarkable for somnolence, asterixis, and periumbilical ecchymoses. His initial lab results showed markedly elevated liver function tests, prolonged coagulation studies, and a urine drug screen that was positive for cocaine. During the patient's interview, his vital signs became unstable. He was intubated for airway protection and transferred to a tertiary care facility for liver transplant evaluation with the diagnosis of cocaine-induced acute liver failure. There he received NAC, lactulose, rifaximin, and vasopressors. On day two of treatment, his clinical condition greatly

improved, and he was extubated. He continued to receive NAC until day five when his liver function tests and coagulopathy improved enough to stop treatment. This case report highlights the clinical benefit of NAC in a case of cocaine-induced acute liver failure, improving the patient's survival and eliminating his need for a liver transplant.

Fluorofenidone protects against acute liver failure in mice by regulating MKK4/JNK pathway.

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Acute liver failure (ALF) is a life-threatening disease characterized by abrupt and extensive hepatic necrosis and apoptosis, resulting in high mortality. The approved drug, N-acetylcysteine (NAC), is only effective for acetaminophen (APAP)-associated ALF at the early stage. Thus, we investigate whether fluorofenidone (AKF-PD), a novel antifibrosis pyridone agent, protects against ALF in mice and explore its underlying mechanisms. ALF mouse models were established using APAP or lipopolysaccharide/D-galactosamine (LPS/D-Gal). Anisomycin and SP600125 were used as JNK activator and inhibitor, respectively, and NAC served as a positive control. Mouse hepatic cell line AML12 and primary mouse hepatocytes were used for in vitro studies. AKF-PD pretreatment alleviated APAP-induced ALF with decreased necrosis, apoptosis, reactive oxygen species (ROS) markers, and mitochondrial permeability transition in liver. Additionally, AKF-PD alleviated mitochondrial ROS stimulated by APAP in AML12 cells. RNA-sequencing in the liver and subsequent gene set enrichment analysis showed that AKF-PD significantly impacted MAPK and IL-17 pathway. In vitro and in vivo studies demonstrated that AKF-PD inhibited APAP-induced phosphorylation of MKK4/JNK, while SP600125 only inhibited JNK phosphorylation. The protective effect of AKF-PD was abolished by anisomycin. Similarly, AKF-PD pretreatment abolished hepatotoxicity caused by LPS/D-Gal, decreased ROS levels, and diminished inflammation. Furthermore, unlike NAC, AKF-PD, inhibited the phosphorylation of MKK4 and JNK upon pretreatment, and improved survival in cases of LPS/D-Gal-induced mortality with delayed dosing. In summary, AKF-PD can protect against ALF caused by APAP or LPS/D-Gal, in part, via regulating MKK4/JNK pathway. AKF-PD might be a novel candidate drug for ALF.

Albumin fusion prolongs the antioxidant and anti-inflammatory activities of thioredoxin in mice with acetaminophen-induced hepatitis.

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Overdoses of acetaminophen (APAP) are a major cause of acute liver failure. N-Acetylcysteine (NAC) is the standard therapy for patients with such an overdose because oxidative stress plays an important role in the pathogenesis of APAP-induced hepatitis. However, NAC is not sufficiently efficacious. We previously developed a recombinant human serum albumin (HSA)-thioredoxin 1 (Trx) fusion protein (HSA-Trx), designed to overcome the unfavorable pharmacokinetic and short pharmacological properties of Trx, an endogenous protein with antioxidative and anti-inflammatory properties. In this study, we investigated the therapeutic impact of HSA-Trx in mice with APAP-induced hepatitis. The systemic administration of HSA-Trx significantly improved the survival rate of mice treated with a lethal dose of APAP compared with saline. HSA-Trx strongly attenuated plasma transaminases in APAP-induced hepatitis mice compared with HSA or Trx, components of the fusion protein. HSA-Trx also markedly caused a diminution in the

histopathological features of hepatic injuries and the number of apoptosis-positive hepatic cells. In addition, an evaluation of oxidative stress markers and plasma cytokine and chemokine levels clearly showed that HSA-Trx significantly improved the breakdown of hepatic redox conditions and inflammation caused by the APAP treatment. HSA-Trx also significantly decreased oxidative and nitrosative/nitrative stress induced by SIN-1 in vitro. Finally, HSA-Trx, but not the NAC treatment at 4 h after APAP injection, significantly inhibited the elevation in plasma transaminase levels. In conclusion, the findings suggest that HSA-Trx has considerable potential for use as a novel therapeutic agent for APAP-induced hepatitis, due to its long-lasting antioxidative and anti-inflammatory effects.
